



Formation of bubbles at submerged orifices – Experimental investigation and theoretical prediction

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ABSTRACT

The present work reports an experimental investigation on bubble release through submerged orifices. Bubble frequency has been measured as a function of gas flow rate for three different orifice sizes at various pool heights. Needle type conductivity probe has been used for bubble count. Analysis of probe signal not only gives the bubble frequency but also indicates a transition from bubbling to jetting regime. Further, to validate the experimental observations a simple mechanistic model has been developed considering the evolution of non-spherical bubbles at the orifice mouth. Reasonable agreement between the model prediction and the experimental result has been observed.

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1. Introduction

Release of gas bubbles through submerged apertures comprises a complex interplay of different fluid dynamic processes. This phenomenon is encountered in various process equipments like gas–liquid contractor, gas–liquid separator, bubble absorber, bubble column, fermentator etc. Interestingly, release of gas bubbles from a submerged orifice also resembles nucleate boiling. A study of the former is expected to throw some light on formation and departure of bubbles, waiting period and inception of jetting which are common for nucleate boiling.

It is not surprising that the intricate process of bubble evolution at a submerged orifice has motivated a large volume of research activities over the decades. One can trace a continuous development of the topic through experimental investigations, analytical modelling and computational simulations. In a recent review Kulkarni and Joshi [1] has discussed this phenomena in the most comprehensive manner. Nevertheless, some of the important investigations on this topic are briefly discussed in this section. Early experiments were conducted by Davidson [2], Walters and Davidson [3] and McCann and Prince [4]. Mainly the shape, the volume and the frequency of the released bubbles were estimated for a variation of different operating parameters. Effect of chamber volume was investigated by Marmur and Rubin [5] while Akagi

et al. [6] and Che and Chen [7] carried out some experiments to understand the weeping rate as well as the effect of chamber volume on the process of bubbling. Kumar and Kuloor [8] and Terasaka and Tsuge [9] reported some detailed review of the previous experimental studies. But all these experiments depend on the visual observation for calculation of bubble frequency. As the individual pattern of the bubble period is not monitored the qualitative comparison of experimental data for the development of basic understanding of the process was not possible.

McCann and Prince [3], Kupferberg and Jameson [10] and LaNauze and Harris [11] attempted to analyze the process based on simplified assumptions. But most of these models are very case sensitive and treat the bubble as a gaseous spherical lump. Marmur and Rubin [12] relaxed the limitation of spherical growth considering the local motion of the gas liquid interface and introduced the concept of “added mass”. Their model can also predict the transition from characteristics of single bubble to bubble jetting region. Subsequently, Tan and Harris [13], Pinczewski [14], Terasaka and Tsuge [9] proposed different models dividing the bubble interface into a number of elements (interfacial element method). Though these models accounted the growth of non-spherical bubble, the mechanism of various bubble behaviors at submerged orifices remains far from fully understood.

The model of Zhang and Tan [15] applied the potential flow theory to describe the bubble formation, its growth and departure. They used Oseen’s modification [16] to potential flow to propose detachment criteria for a growing bubble. Liquid weeping and wake pressure due to departing bubble is accounted in their model

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Nomenclature

a	instantaneous bubble radius for submerged orifice (m)	R_1	first radius of curvature of the growing bubble (m)
a_c	laplace constant	R_2	second radius of curvature of the growing bubble (m)
b	thickness of the plate (m)	s	perpendicular distance between bubble centre and orifice (m)
B	bond number	t	time (s)
C_g	constant for Eq. (8)	t_w	waiting period (s)
C_p	constant pressure specific heat (kJ (kg K)^{-1})	T	temperature (K)
C_v	constant volume specific heat (kJ (kg K)^{-1})	u	transformed coordinate
f	fanning friction factor (dimensionless)	U	bubble rising velocity (m s^{-1})
Fr	Froude number	u_l	liquid velocity for the refill stage (m s^{-1})
g	acceleration due to gravity (m s^{-2})	V	instantaneous voltage (volt)
Ga	Galileo number	V_b	instantaneous bubble volume (m^3)
h_1	enthalpy of the air entering at the chamber (J (kg K)^{-1})	V_c	chamber volume (m^3)
h_0	height of the pool above the surface or orifice (m)	$V_{\max}$	maximum value of voltage (volt)
k_b	factor stated in Eq. (7)	$V_{\min}$	minimum value of voltage (volt)
m_{in}	mass flow rate into the tank below the submerged orifice (kg s^{-1})	V^*	normalized voltage (dimensionless)
M_{ch}	mass of the air inside the chamber (kg)	x	coordinate axis
P_b	bubble pressure (Pa)	y	coordinate axis
P_c	chamber pressure (Pa)	z	transformed coordinate
P_g	gas pressure (Pa)	z_1	static height of air inlet point from datum (m)
P_h	hydrostatic pressure (Pa)		
P_σ	capillary pressure (Pa)		
P_∞	ambient pressure (Pa)		
q	mass flow rate into the bubble through orifice (kg s^{-1})		
Q	mass flow rate into the chamber below orifice (kg s^{-1})		
r_0	orifice radius (m)		
R	universal gas constant ($\text{kJ kg}^{-1} \text{K}^{-1}$)		
R_0	radius of curvature at the apex of the bubble (m)		

Greek letters

ρ_0	density of incoming air to the chamber (kg m^{-3})
ρ_{int}	initial density of air inside the chamber (kg m^{-3})
ρ_w	water density (kg m^{-3})
σ	surface tension (N m^{-1})
γ	ratio of fluid's specific heats

to predict the motion of a spherical bubble. Xiao and Tan [17] used boundary integral method to predict the bubble shape, chamber pressure below the orifice and frequency for a single submerged orifice assuming the flow to be irrotational. They also showed the effect of surface tension and liquid circulation around the growing bubble. Recently Gerlach et al. [18] investigated quasi static condition of non-spherical bubble formation using Laplace equation and reported that two types of bubble formation occurs depending on the Young contact angle, which is defined by the solid gas and liquid contact point. In the first case bubble base remains attached to the orifice rim and in the other it spreads along the surface. Recently, Das and Das [19] employed Smoothed Particle Hydrodynamics for simulating the process and could obtain very good agreement of the simulation results with the experimental data taken from different sources.

The present work aims to introduce a new methodology for measuring the frequency of the formation and release of bubbles from submerged orifices using air and water as the test fluids. A conductivity probe has been used for estimating the bubble count. The analysis of the probe signal provides a better method to understand the dynamics of the bubbling process. A simple analytical model has been employed to predict the experimental bubble frequency. Reasonable agreement between the model and the experimental data is observed.

2. Experimental investigation

The experimental facility is schematically shown in Fig. 1. Air and distilled water are used as test fluids. Air is supplied at a constant pressure (2.5 bar absolute at the delivery of the compressor) by a small oil free compressor (maximum allowable working pressure 5 bar, 350 rpm) through a orifice submerged in stagnant water pool. The air passes through a series of inline and bypass needle

valves that helps to control the flow rate. For the present range of experiments nearly 50% of the intake mass is bypassed. Air first enters in a leak proof (except the orifice above the tank) cylindrical tank of 200 mm diameter and 150 mm length; this volume is very large in comparison to the bubble departure volume. The tank is fabricated by 5 mm thick brass plate and clamped properly to withstand the thrust due to the air entry. The tank is provided to dampen the entry turbulences created by the air. At the top of the air chamber small circular orifices are made at the centre of a 5 mm thick brass plate. The employed orifice diameters are 1, 2 and 3 mm. The air volume flow rate is regulated from 0.2 to 8.0 ml/s by the valve system. Air after release from the orifice enters in a rectangular tank. The cross section of the tank above the orifice is 200 mm $\times$ 200 mm. The maximum height of the liquid level in the water tank is 200 mm. Proper care has been taken to ensure that air escape to the water tank only through the orifice. Other joints are sealed with gaskets, o rings and sealants. The design of the tank above the orifice has been made judiciously based on the requirements of the present study. The size of the tank is such that the escaping bubbles will not feel the "wall effect". Tank walls are provided with flat glass windows to facilitate visualization. Orifice section is kept projected up by 30 mm in the water chamber to avoid any visual obstruction from the metallic walls of the water tank. Additionally, flat walls reduce the optical distortion. The tank was filled with distilled water. Five different heights, 70 mm, 90 mm, 110 mm, 130 mm and 150 mm of water above the orifice have been considered for the present study. The top of the rectangular tank is open to atmosphere. During the flow of air through the orifice, small amount of water also seeps into the air tank. Care has been taken to maintain a constant height of the water pool. The measurement of gas flow rate is most challenging as the flow rate is very small and is intermittent through the orifice. A conical funnel is placed inside the water pool above the orifice. Air emerges as a bubble inside the water pool and after detachment gets

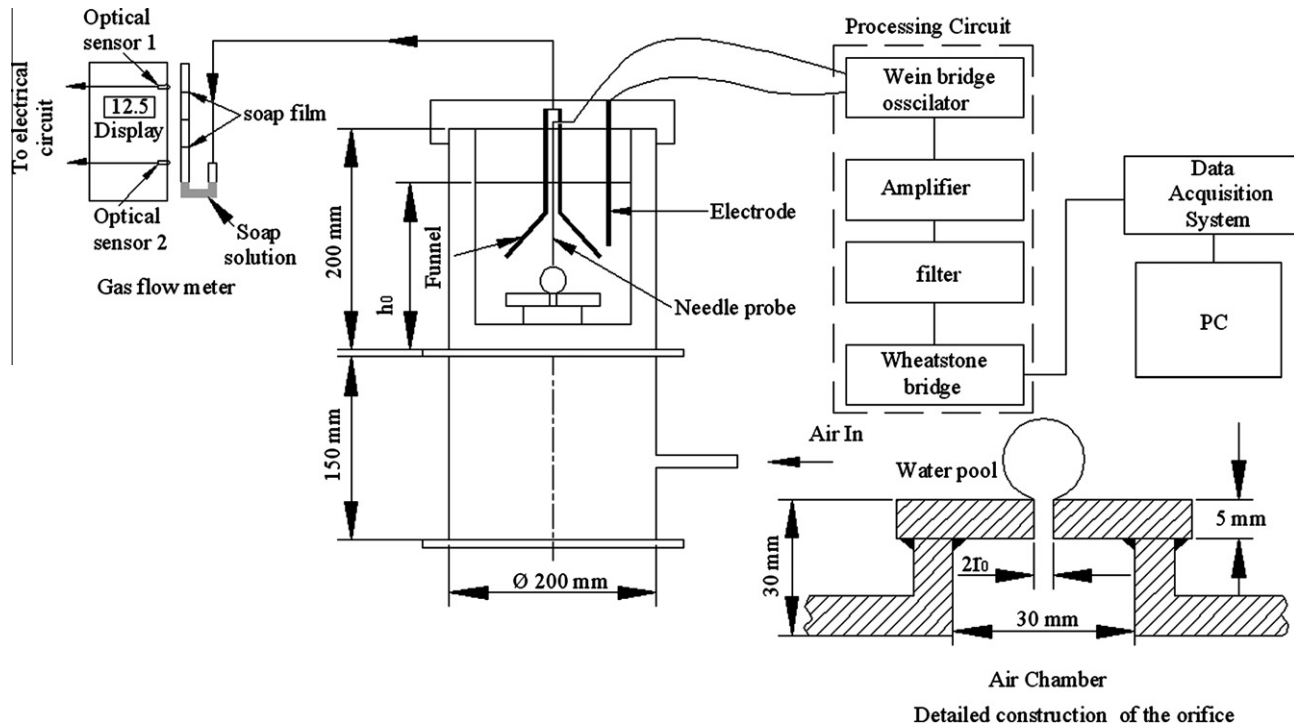


Fig. 1. Schematic diagram of experimental set up.

entrapped inside the funnel. The funnel guides the air bubble towards a micro-controller based digital flow meter which works on the principle of soap bubble rise. It uses optical (IR) sensors and can measure low flow rates with a resolution of 0.1 ml/s with an uncertainty of $\pm 2\%$ at the highest flow rate (8.0 ml/s). Care has been taken before preparing the dilute soap solution so that at high flow rate the soap film does not break. Volume flow rate is calibrated using the velocity of the soap foam determined by the sensors and displayed during the experiment. A schematic diagram of the method for measuring the air flow rate is shown in the Fig. 1.

3. Measurement of bubble frequency

A needle type conductivity probe is used to estimate the bubble frequency. The probe assembly has two electrodes namely needle electrode and large electrode. A 2 mm diameter stainless steel wire has been selected for the probe construction from the consideration of rigidity. By precision grinding, the diameter of the wire has been reduced towards the free end and a narrow angle cone has been provided at the tip. The entire length of the probe is electrically insulated apart from 0.5 mm at the tip. The needle electrode is inserted through the funnel and its tip is suspended immediately (10 mm) above the orifice. A copper strip forms the other electrode and is immersed in the water pool away from the orifice. The needle probe is made using a fine steel wire. It is properly insulated except the probe tip and can detect a bubble just after its departure from the orifice mouth. The two electrodes are connected to a processing circuit which converts the change of resistance between them into a voltage signal. It consists of Wein bridge oscillator, amplifier, filter and one balanced Wheatstone bridge. As the bubbles are passed through the probe tip, the in situ resistances are changed which makes the Wheatstone bridge unbalanced. Finally, the voltage–time signal generated by Wheatstone bridge can be displayed and stored in a PC through a data acquisition system. The response time of the processing circuit along with the data acquisition system is quite low and of

the order of 10^{-2} s. At maximum flow rate (8.0 ml/s) the uncertainty of measuring bubble frequency is $\pm 2\%$. For lower flow rates the error in measurement of bubble frequency is quite low as the response time of the processing circuit is small. The bubble frequency is measured in Hz.

4. Experimental results

At low air flow rates, bubbly flow with discrete spherical bubbles can be seen. With the increase of flow rate, frequency of the bubble departure increases and at a higher flow rate it is really difficult to distinguish one bubble from the other. During fast bubbling through the orifice the visual bubble count may lead to errors. As the bubble at the orifice mouth coalesces with the preceding bubble right at its top one can say that air jetting takes place. The phenomena is analyzed from the probe signals and their probability distribution function (PDFs). Though the experiments were conducted for different flow rate and liquid pool height for a single orifice diameter, only few representative results will be depicted to avoid the repetition. Probe signals for three different flow rates are shown in Fig. 2 to qualitatively show the change of bubble departure frequency for 1 mm orifice diameter. The downward spikes in the voltage signal indicate the presence of a bubble. Probability distribution functions are also shown for those flow rates.

To avoid the arbitrariness and for ease of comparison, voltage signal is normalized as V^* for constructing the PDF curves. Where,

$$V^* = \frac{V - V_{\min}}{V_{\max} - V_{\min}} \quad (1)$$

Hence in normalized voltage scale, liquid contact occurs at $V^* = 1$ and air contact makes it $V^* = 0$. One well defined peak near $V^* = 1$ indicating a very high probability of the existence of liquid phase at the probe tip is depicted in the PDF curves (Fig. 3). But another minor peak nearly at $V^* = 0.15$ is shown and it is increasing its magnitude with increase in flow rate. This peak is created due to contact of the emerged bubble with the needle probe. It can

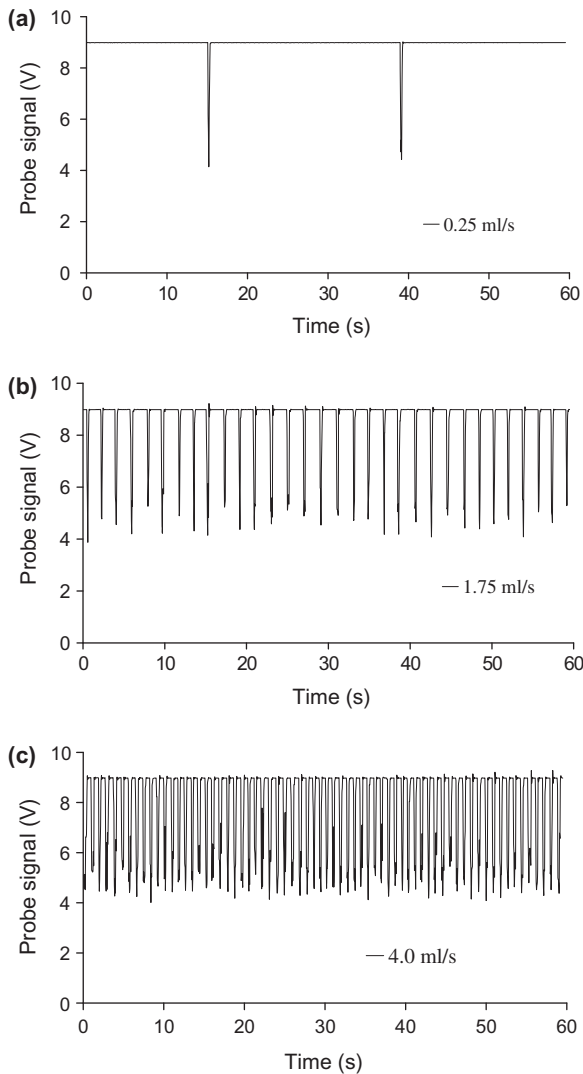


Fig. 2. Probe signal for bubble formation through 1 mm orifice diameter at 110 mm pool height.

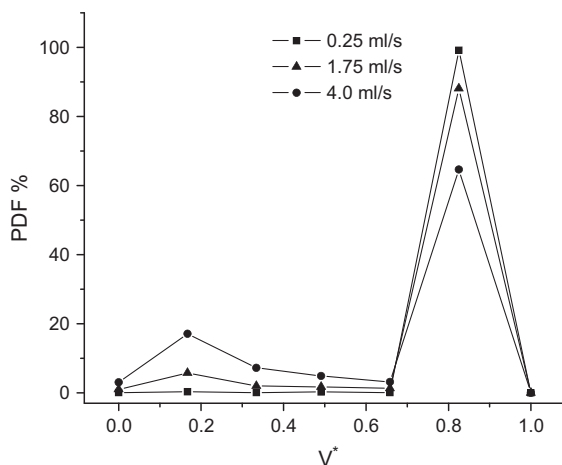


Fig. 3. Probability distribution function for probe signals of different flow rates.

be concluded from the PDFs that bubble frequency increases with the increase of the flow rate. But the pattern of the signal is highly complex at high flow rate and indicates that the phenomenon can lead to a chaotic situation if flow rate is increased further.

Tests were carried out for various pool heights as well. Probe signals for 1 mm orifice diameter and two different pool heights (70 mm and 150 mm) are shown in Figs. 4a and 4b. It can be seen that as the pool height above the submerged orifice increases, the bubble departure frequency decreases. The trend is same for all pool heights and orifice diameters. However, the change in bubble frequency is more significant for large flow rates. From the probe signal it is clear that the process of bubbling changes significantly as the pool height varies. The randomness of the signals decreases with the increase of the pool height. Fig. 5 gives a consolidated view how the bubble frequency changes with flow rate for different pool heights.

The effect of orifice diameter on bubble frequency can be noted from Figs. 6a and 6b. For 1 mm orifice diameter, 27 bubbles depart within a span of 1 min whereas, for 3 mm pool height 18 bubbles depart during the same time span. Comparisons for the other flow rates are shown for two different orifice diameters in Fig. 7. It is obvious that the bubble diameters for 1 mm orifice are smaller than those for 3 mm orifice as a result bubble frequency for higher orifice diameter decreases. As a bubble leaves the orifice mouth a small amount of liquid also seeps into the air chamber below the orifice. This phenomenon is addressed as weeping and has been addressed by a number of researchers [6,7,19]. In the present study though the phenomenon of weeping has not been investigated rigorously it has been monitored for all the experimental runs for a complete assessment. It was observed that the release of a larger bubble also resulted in the down flow of a larger liquid mass through the orifice. Water accumulated due to weeping was collected from the tank after every 30 min and measured. For 1 mm orifice weeping rate was nearly 0.02 ml/s whereas, it was 1.0 ml/s in case of 3 mm orifice.

To understand the dynamics of gas release through the submerged orifice Fast Fourier transform (FFT) of the probe signal are performed. For a typical test case of 1 mm orifice diameter and 70 mm pool height FFT signals obtained for increasing air flow rates are shown in Fig. 8. At low air flow rate bubbles are released at a regular interval. Corresponding FFT signals indicate dominant peaks (Fig. 8a–c). In any of these figures the progressively decreasing peaks at regular intervals indicate their occurrence at integral multiple of the characteristic harmonic. With the increase of air flow rate air jetting occurs at the orifice mouth in the place of periodic bubble rise. The phenomenon is characterized by fluctuating motion of the air jet. Accordingly the probe signal depicts a relatively random nature. The FFT signal also represents this random phenomenon. With the increase of air flow rate the number and the height of the dominant peaks reduce. Ultimately a single peak is observed in Fig. 8f. Though the peak height is lower compared to that observed in Fig. 8b the area under the peak is much larger indicating the existence of many other sub frequencies. In other word, the signal represents the randomness of the jetting phenomena.

The transition between the bubbling to jetting regime has been carefully examined for various pool heights and orifice diameters. It is reported in Table 1. It can be concluded that higher air flow rate is needed for a transition from bubbling to jetting as the orifice diameter increases.

5. Model development

It is needless to mention that the formation, growth and departure of bubbles at submerged orifice depend on a large number of geometrical parameters, fluid properties and operating conditions. In such a scenario it is always convenient to have a correlation preferably in terms of nondimensional numbers. This not only reduces the number of variable but also provides a generic picture of

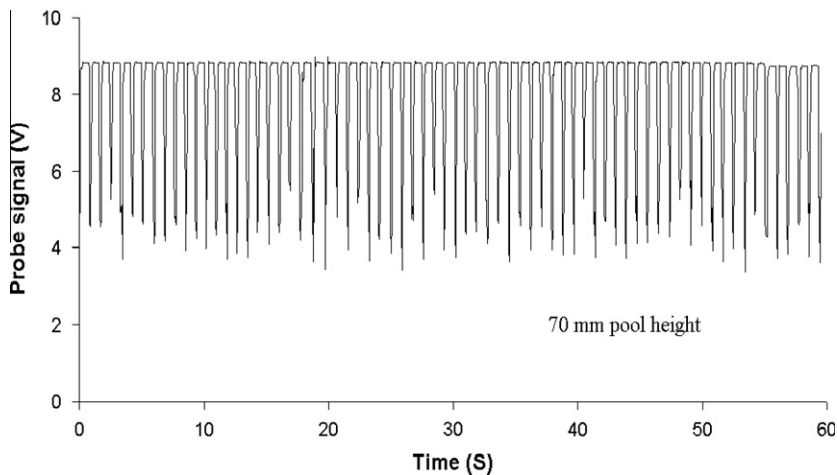


Fig. 4a. Probe signal for 1 mm orifice diameter and 70 mm pool height at 3.0 ml/s flow rate.

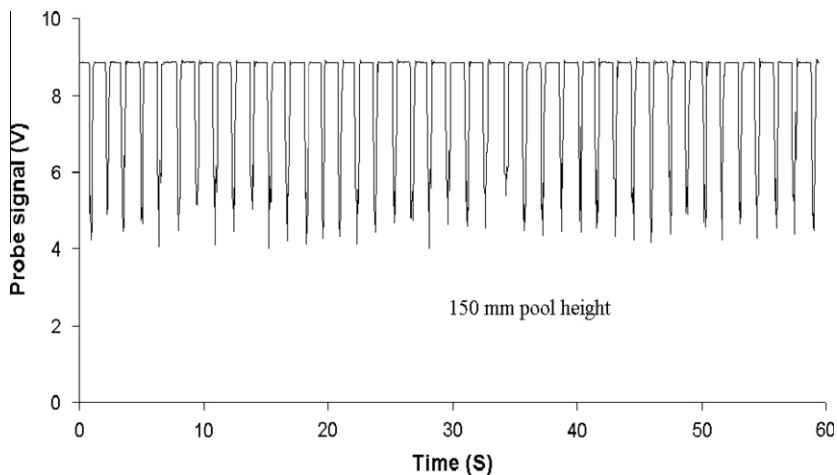


Fig. 4b. Probe signal for 1 mm orifice diameter and 150 mm pool height at 3.0 ml/s flow rate.

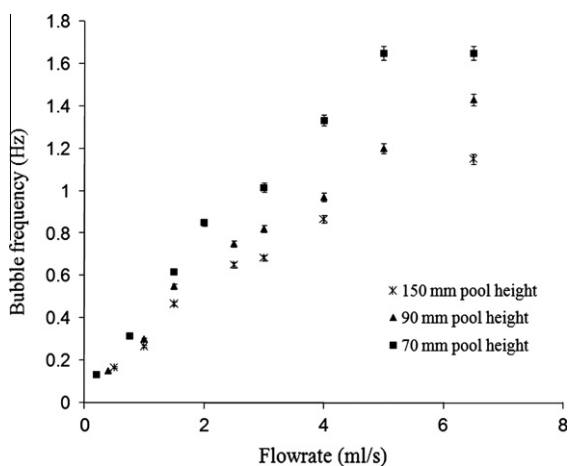


Fig. 5. Bubble frequency at three different pool heights for 1 mm orifice diameter.

the complex physical phenomena. A large number of correlations relating departing bubble diameter with the fluid properties and process parameters are available in literature. Jamialahmadi et al. [20] has given a concise account of such correlations. Jamialahmadi et al. [20] further developed another correlation using three impor-

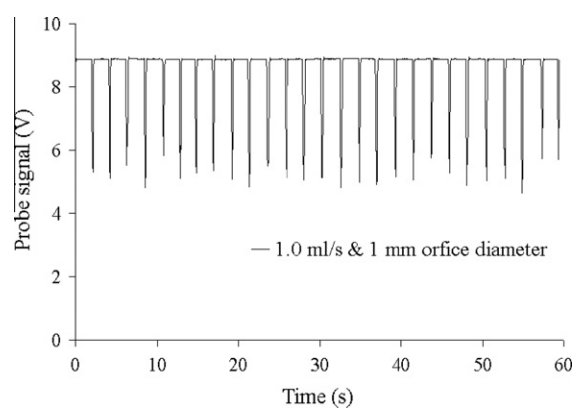


Fig. 6a. Probe signal for 1 mm orifice diameter at 1.0 ml/s flow rate and 90 mm pool height.

tant nondimensional numbers namely Bond number (Bo), Galileo number (Ga) and Froude number (Fr) and use the same to validate the experimental data covering a wide range. They have observed the correlation to outperform all other existing correlations. Recently, Gerlach et al. [18] compared their computational predictions against this correlation and obtained excellent agreement. However, one can point out two important limitations of this

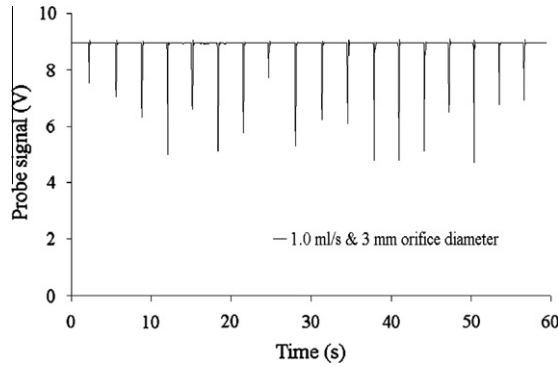


Fig. 6b. Probe signal for 3 mm orifice diameter at 1.0 ml/s flow rate and 90 mm pool height.

correlation. Firstly the correlation does not account for the influence of liquid height over the orifice on the process of bubbling. Secondly, the correlation predicts the bubble diameter. This essentially assumes a spherical bubble. The formation of non-spherical bubble has been established by a number of experimental studies [4,21] as well as theoretical modelling [17,22]. This has motivated us to develop a simple and computationally inexpensive model which will take care of the non-sphericity of the bubble and the influence of the liquid height above it.

The system under consideration is idealized as shown in Fig. 9. It consists of two chambers separated by a thin plate. The thin plate has an orifice at its centre. The lower chamber termed as air chamber has a volume V_c and gets the air supply through a side

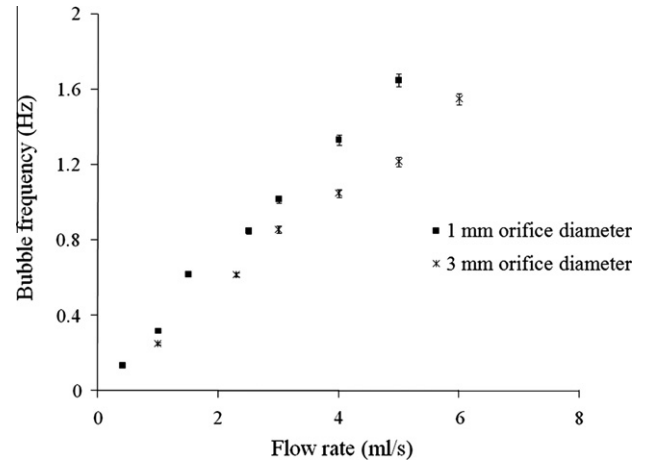


Fig. 7. Frequency comparison for two different orifice diameters at 1.0 ml/s flow rate and 90 mm pool height.

tube. The upper chamber is the liquid chamber and is open to atmosphere at the top. High pressure air is injected into the pool through the orifice from the air chamber. The main assumptions involved in the development of the theory are as follows:

- I. The gas phase is compressible and obeys ideal gas law.
- II. Volume of liquid around the bubble is very large compared to the volume of bubble.
- III. The formation of bubble is unaffected by the presence of the previous bubble.

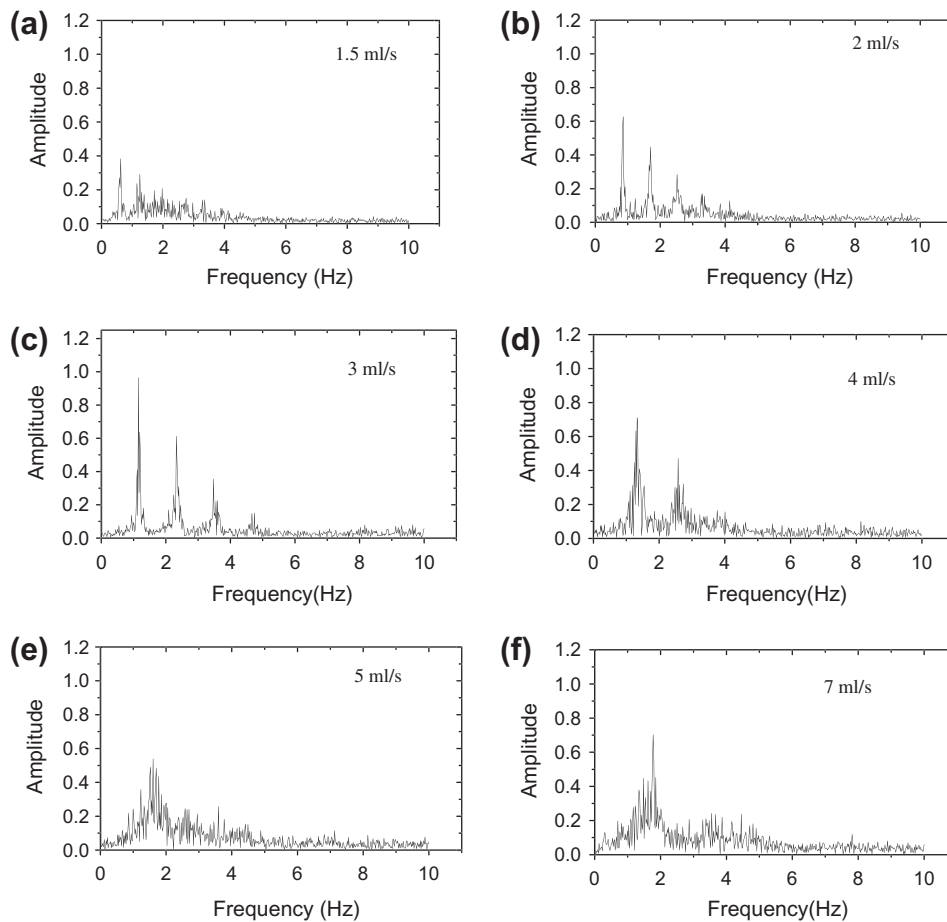


Fig. 8. FFT of bubble frequency for different flow rate at 70 mm pool height and 1 mm orifice diameter.

- IV. The pressure variation inside the bubble is neglected.
- V. The simulation is done only up to the instant of bubble release.
- VI. Bubbles are non-spherical but axially symmetric.
- VII. Other assumptions will be mentioned wherever necessary.

The simulation is made of two parts. In the first part a thermodynamic model of the entire process [17] is done considering the conservation of mass and energy. As an outcome of the model mass flow rate through the orifice can be estimated at every instant. This gives the rate of volume change of the bubble. In the second part the details of bubble growth and bubble shape evolution is modeled. The evolution cycle of the bubble can again be divided into two parts. Initially the pressure inside the chamber is lower than that of outside the orifice as bubble has to overcome the pressure of the liquid column over it. Time required to overcome this pressure is called waiting period (t_w) and bubble nucleation starts after this period. At the end of waiting period the air pressure at the orifice mouth becomes less or equal to the air pressure at the chamber. Due to the favorable pressure gradient air embryo protrudes in the water pool causing bubble nucleation and start of the bubble period. Taking the air chamber as a control volume and applying first law of thermodynamics one gets:

$$m_{in} \left(h_1 + \frac{v_1^2}{2} + gz_1 \right) = \frac{\partial}{\partial t} (M_{ch} C_v T) \quad (2)$$

From Eq. (2) we can calculate the waiting time before bubble nucleation

$$t_w = \left(\frac{P_\infty + \rho_w g h_0}{RT} - \rho_{int} \right) \frac{V_c C_v T}{A} \quad (3)$$

where

$$A = \rho_0 Q \left(C_p T + \frac{v_1^2}{2} + gz_1 \right) \quad (4)$$

After that nucleation starts and non-spherical bubble embryo protrude through the orifice.

Following Zhang and Tan [15] an energy balance for the bubble can be written and neglecting the heat transfer and finally the following relationship can be obtained:

$$V_c P_c = \gamma P_c (Q - q) \quad (5)$$

where

$$q = \frac{dV_b}{dt} \quad (6)$$

Inflow of gas into the bubble at any instant depends on the difference of the chamber and bubble pressure:

$$q = k_b (P_c - P_b)^{0.5} \quad (7)$$

Here k_b is a factor and it is reported by Miyahara and Takahashi [23] as stated below:

Table 1
Experimental observation for jetting phenomena.

Orifice diameter (mm)	Pool height (mm)	Flow rate at which jetting starts (ml/s)
1	70	4.0
	90	4.5
	150	5.0
2	70	5.5
	90	6.5
	150	7.0
3	70	6.5
	90	7.0
	150	8.0

$$k_b = \pi r_0^2 \sqrt{2/\rho_g C_g} \quad \text{where} \quad C_g = 1.5 + 2fb/r_0 \quad (8)$$

Here b is the orifice plate thickness and f accounts for the frictional loss.

Instantaneous pressure inside the bubble having radius “ a ” is governed by the modified Rayleigh equation as follows [4]:

$$P_b = \rho_w \left(\frac{3}{2} \dot{a}^2 + a \ddot{a} - gs \right) - \frac{\rho_w U^2 a}{4(s+a)} \left(1 + \frac{5s}{2a} - \frac{3s^3}{2a^3} \right) + P_\infty + \frac{2\sigma}{a} \quad \text{for } s < a \quad (9a)$$

$$\text{and for } s > a \quad P_b = \rho_w \left(\frac{3}{2} \dot{a}^2 + a \ddot{a} - gs \right) - \frac{\rho_w U^2}{4} + P_\infty + \frac{2\sigma}{a} \quad (9b)$$

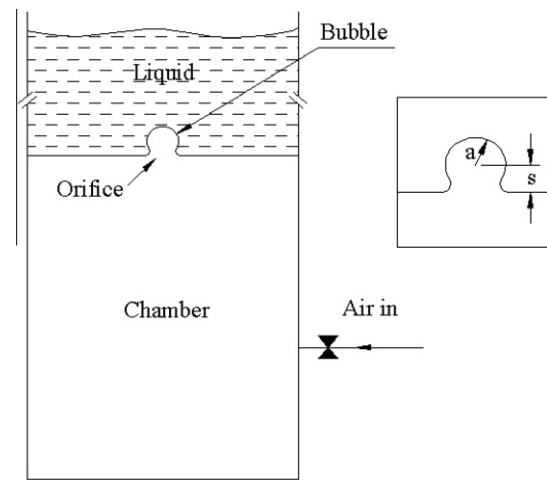


Fig. 9. Schematic diagram of the physical system.

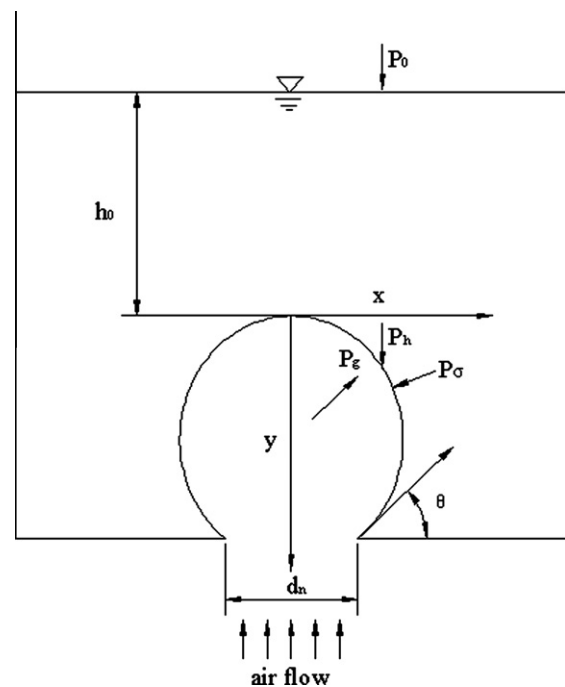


Fig. 10. Local force balance inside the bubble.

Here, s is the distance of centre of gravity of the bubble from the orifice mouth (Fig. 9).

Balancing the inertia and buoyancy force of the injected mass at any instant one gets:

$$\frac{d}{dt} \left(\frac{11}{16} \rho_w V_b U \right) = (\rho_w - \rho) V_b g \quad (10)$$

Bubble rising velocity U can be defined as the velocity of the bubble centre and it is assumed to be

$$U = \frac{ds}{dt} \quad (11)$$

Initial conditions are : $P_c(0) = P_b(0) = P_\infty + 2\sigma/r_0$

$$q(0) = U(0) = s(0) = 0 \quad (12)$$

$$a(0) = r_0$$

Using Eqs. (5)–(8), (9a), (9b), (10), (11) and the initial conditions instantaneous air injection to the bubble can be evaluated.

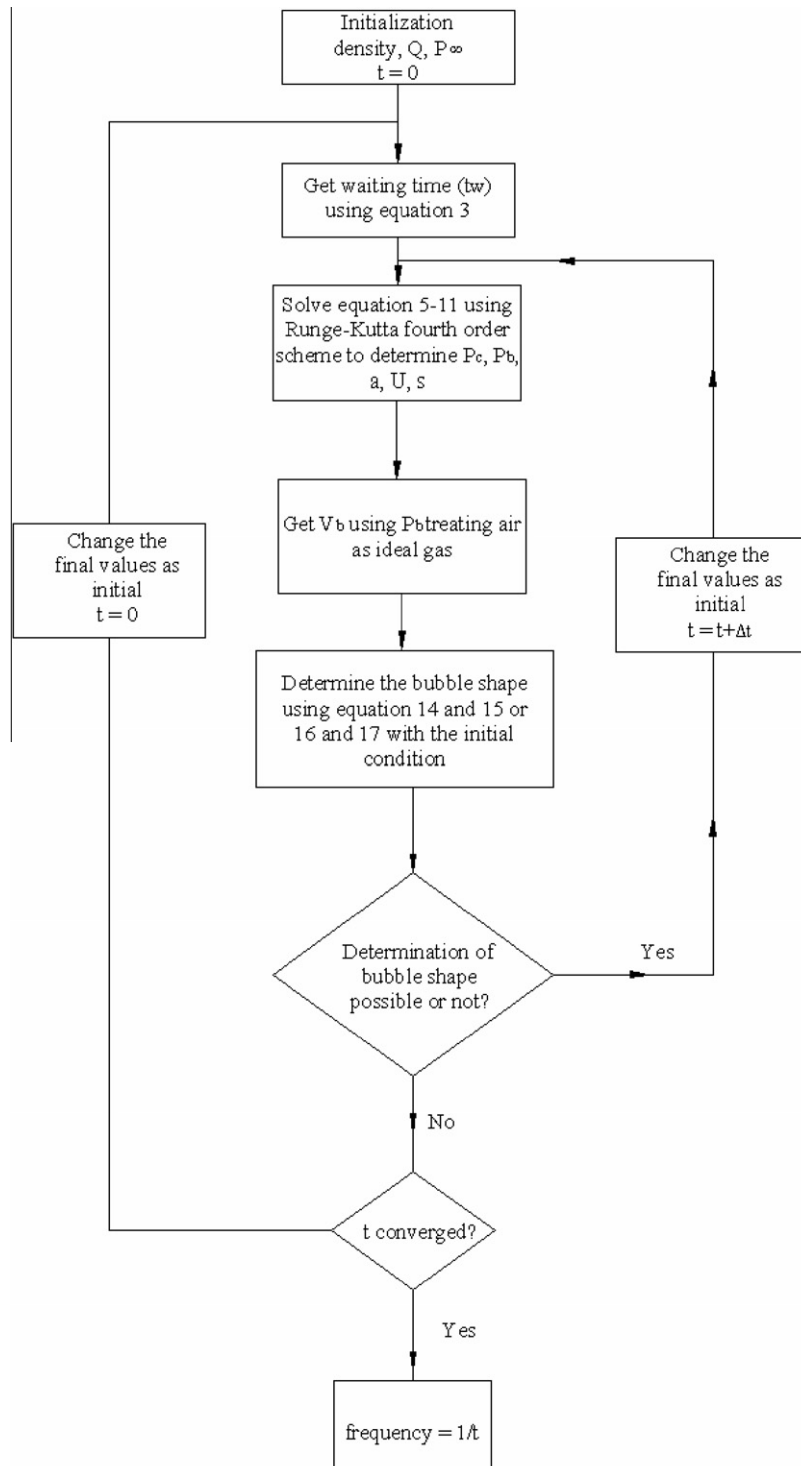


Fig. 11. Flow chart for the calculation procedure.

Assuming the quasi static growth of the bubble and following the procedure of Gerlach et al. [18] the evolution of bubble shape has been predicted. Origin of the coordinate is taken at the vertex of the bubble and y coordinate is pointing towards the nozzle. As the bubble is symmetric with respect to y , x coordinate can be taken in any side of the bubble. Gas pressure (P_g), capillary pressure (P_σ), hydrostatic pressure (P_h) is equated to the ambient pressure (P_∞) to determine the resultant pressure for any coordinate position as shown in Fig. 10

$$P_g = P_\sigma + P_\infty + P_h = \sigma \left(\frac{1}{R_1} + \frac{1}{R_2} \right) + P_\infty + \rho_w g(h_0 + y) \quad (13)$$

Using the general relationship for the radius of curvatures Gerlach et al. [18] obtained the following first order equations for the bubble shape:

$$\frac{dz}{dx} = -\frac{z}{x} + 2 \left(\frac{1}{R_0} - y \right) \quad (14)$$

$$\frac{dy}{dx} = \frac{z}{(1 - z^2)^{1/2}} \quad (15)$$

with the initial condition at $x = 0$, $y = 0$ and $z = 0$

Starting from $x = 0$ forward marching in x direction has been done. But the bubble has a point of inflexion $dy/dx \approx 1.0$ after which Eqs. (14) and (15) are not valid. It can be taken care by substituting $u = (1 - z^2)^{1/2}$ after reaching $dy/dx = 1.0$. This change Eqs. (14) and (15) in the following manner:

$$\frac{du}{dy} = \frac{(1 - u^2)^{1/2}}{x} - 2 \left(\frac{1}{R_0} - y \right) \quad (16)$$

$$\frac{dx}{dy} = \frac{u}{(1 - u^2)^{1/2}} \quad (17)$$

For these two equations the final values of Eqs. (14) and (15) are taken as initial conditions and march forward in y direction. Marching should be continued till the x coordinate reaches the radius of the orifice. Bubble shape is determined for each time step and air injection (Eqs. (5)–(8), (9a), (9b), (10), (11)).

Departure of a bubble imposes a singularity to its growth phase. At that instant, no physically possible value of R_0 can produce a bubble with a closed contour satisfying the corresponding volume. Any perturbation or a continuation of the flow through the orifice causes the bubble to leave the orifice opening. This gives the maximum possible bubble volume and indicates the termination of its growth. Corresponding to this criterion, bubble departure diameter can be estimated for any fixed flow rate, orifice size and pool height combination.

6. Solution procedure

Eqs. (5)–(8), (9a), (9b), (10), (11) are simultaneously solved for P_c , P_b , a , u , s using Runge–Kutta fourth order scheme respectively. Initial conditions are stated in Eq. (12). At each time step bubble shape is calculated using Eqs. (14)–(17) depending on the value of dy/dx . Simulations were done until the shape cannot be determined by the above equations which prompt the bubble to depart. Calculations are carried out further for the next bubble until the bubble departure time converges. It has been seen that the departure time usually converges for the third bubble. Later the departure time becomes quite identical. A flow chart has been given in Fig. 11 for the solution procedure.

7. Comparison with experiment

Based on the above theory, the formation, growth and departure of the bubbles are simulated within a large gas flow rate regime from 0.5 ml/s to 7.0 ml/s. The comparison of theoretical bubble frequency with experimental data is stated in Fig. 12 for two different orifice diameter and 70 mm pool height. It can be noted that the present model prediction and experimental results are showing similar trend. At low flow rate the model prediction is close to the experimental observation while there is over prediction at high flow rate. The present model neglects the interaction of the surrounding liquid phase with the growing and departing bubble. This may have a significant effect when frequency of bubble release increases. Besides, it is possible that the conductivity probe can miss a few bubbles at high gas flow rate. As a result, a greater mismatch at this condition is not unexpected.

Effect of pool height is also tracked by the present model. Model prediction for three different pool heights in 1 mm orifice diameter is shown in Fig. 13. Though the model predicts the trend of bubble frequency for different pool heights, frequency deviates as the pool height increases. Developed model can be applied for the prediction of bubble frequency from orifice mouth within a range of operating parameter where the bubble cycle is not biased by its preceding and successor bubbles. It may further be noted that the process of bubble formation, its growth and its departure from the submerged orifice is rather complex. On the other hand, the

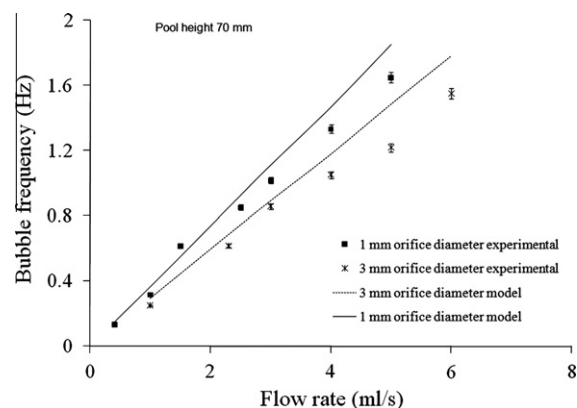


Fig. 12. Comparison of experimental data and model prediction for two orifice diameter and fixed pool height.

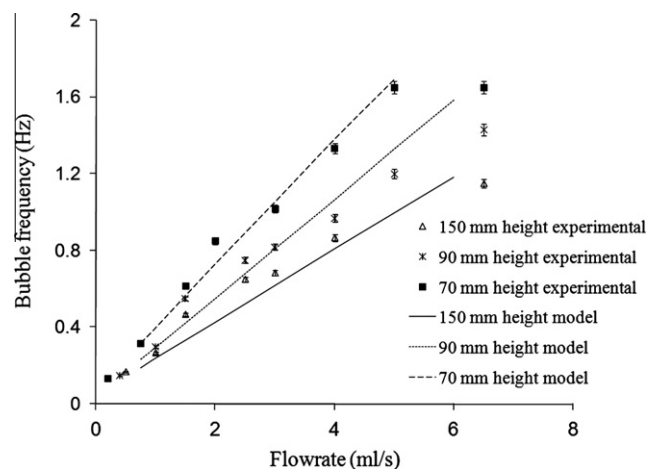


Fig. 13. Comparison of experimental and model prediction for various pool heights.

present model is highly simplified and does not take care of the local hydrodynamics which in turn dictates the bubble dynamics. While, the model can be utilized for a quick design calculation, accurate simulation of the process can only be made through computational fluid dynamics.

8. Conclusion

The present work reports an experimental investigation as well as an analytical model of the process of bubble evolution at submerged orifice. Experimental data on bubble frequency as a function of gas flow rate have been obtained for orifices of three different diameters under various pool heights. Conductivity probe has been used to estimate the bubble count. Probe signal has been analyzed further to detect the transition from bubbling to jetting. The results obtained through signal analysis have been corroborated by visual observation.

Finally, a simple mechanistic model has been used to describe bubble evolution at a submerged orifice. Though it is computationally inexpensive, the model is not restricted only to spherical bubbles and can predict the effect of liquid pool height on the process of bubbling. The model perfectly predicts the trend of the experimental data. However, with further improvements better quantitative prediction by the model cannot be ruled out. Moreover, presented experimental data provides a benchmark against which an improvement of the developed model can be tested.

References

- [1] A.A. Kulkarni, J.B. Joshi, Bubble formation and bubble rise velocity in gas liquid systems: a review, *Ind. Eng. Chem. Res.* 44 (2005) 5873–5931.
- [2] J.F. Davidson, Bubble formation at an orifice in an inviscid liquid, *Trans. Inst. Chem. Eng.* 38 (1960) 335–342.
- [3] J.K. Walters, J.F. Davidson, Bubble formation at an orifice in a viscous liquid, *J. Fluid Mech.* 17 (1963) 321–336.
- [4] D.J. McCann, R.G.H. Prince, Bubble formation and weeping at a submerged orifice, *Chem. Eng. Sci.* 24 (1969) 801–814.
- [5] A. Marmur, E. Rubin, Equilibrium shapes and quasi-static formation of bubbles at submerged orifice, *Chem. Eng. Sci.* 28 (1973) 1455–1464.
- [6] Y. Akagi, K. Okada, K. Kosaka, T. Takahashi, Liquid weeping accompanied by bubble formation at submerged orifice, *Ind. Eng. Chem. Res.* 26 (1987) 1546–1550.
- [7] D.F. Che, J.J.J. Chen, Bubble formation and liquid weeping at an orifice submerged in a liquid, *Chem. Eng. Technol.* 62 (1990) 947–953.
- [8] R. Kumar, N.R. Kuloor, The formation of bubbles and drops, *Adv. Chem. Eng.* 8 (1970) 256–368.
- [9] K. Terasaka, H. Tsuge, Bubble formation at a single orifice in highly viscous liquids, *J. Chem. Eng. Jpn.* 23 (1990) 160–165.
- [10] A. Kupferberg, G.J. Jameson, Bubble formation at a submerged orifice above a gas chamber of finite volume, *Trans. Inst. Chem. Eng.* 47 (1969) 241–250.
- [11] R.D. LaNauze, I.J. Harris, Gas bubble formation at elevated system pressure, *Trans. Inst. Chem. Eng.* 52 (1974) 337–348.
- [12] A. Marmur, E. Rubin, A theoretical model for bubble formation at an orifice submerged in an inviscid liquid, *Chem. Eng. Sci.* 31 (1976) 453–463.
- [13] R.B.H. Tan, I.J. Harris, A model for non-spherical bubble growth at an orifice, *Chem. Eng. Sci.* 41 (1986) 3128–3175.
- [14] W.V. Pinczewski, The formation and growth of bubbles at a submerged orifice, *Chem. Eng. Sci.* 36 (1981) 405–411.
- [15] W. Zhang, R.B.H. Tan, A model for bubble formation and weeping at a submerged orifice, *Chem. Eng. Sci.* 55 (2000) 6243–6250.
- [16] H. Lamb, *Hydrodynamics*, sixth ed., Cambridge University Press, Cambridge, 1945.
- [17] Z. Xiao, R.B.H. Tan, An improved model for bubble formation using the boundary-integral method, *Chem. Eng. Sci.* 60 (2005) 179–186.
- [18] D. Gerlach, G. Biswas, F. Durst, V. Kolobaric, Quasi-static bubble formation on submerged orifices, *Int. J. Heat Mass Transfer* 48 (2005) 425–438.
- [19] A.K. Das, P.K. Das, Bubble evolution through submerged orifice using smoothed particle hydrodynamics: basic formulation and model validation, *Chem. Eng. Sci.* 64 (2009) 2281–2290.
- [20] M. Jamialahmadi, M.R. Zehtaban, H. Müller-Steinhagen, A. Sarrafi, J.M. Smith, Study of bubble formations under constant flow conditions, *Trans. IChemE* 79 (2001) 523–532.
- [21] L. Zhang, M. Shoji, Aperiodic bubble formation from a submerged orifice, *Chem. Eng. Sci.* 56 (2001) 5371–5381.
- [22] V.V. Buwa, D. Gerlach, F. Durst, E. Schlucker, Numerical simulation of bubble formation on submerged orifices: period-1 and period-2 bubbling regime, *Chem. Eng. Sci.* 62 (2007) 7119–7132.
- [23] T. Miyahara, T. Takahashi, Bubble volume in single bubbling regime with weeping at a submerged orifice, *J. Chem. Eng. Jpn.* 17 (6) (1984) 597–602.